
LOSSLESS DATA COMPRESSION TOOL

TAI PROJECT #1 - 2025/26

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ABSTRACT

This report presents a comprehensive study of lossless data compression techniques, culminating in the development of an efficient compression tool for the Algorithmic Information Theory (TAI) course. Our implementation combines Burrows-Wheeler Transform (BWT) preprocessing with adaptive Prediction by Partial Matching (PPM) models and multiple entropy coding techniques including Range Coding and Asymmetric Numeral Systems (rANS). Through iterative optimization across six major versions, we achieved an average compression ratio of 54.73% on benchmark datasets, outperforming bzip2 by 0.20 percentage points and winning on 5 out of 8 test files. The final production system (v5.0) incorporates full pipeline parallelism, achieving 2-5× compression speedup while maintaining competitive ratios. This work demonstrates the effectiveness of combining classical preprocessing transforms with modern entropy coding and parallel processing techniques.

Keywords Data Compression · Burrows-Wheeler Transform · PPM · Range Coding · rANS · Parallel Processing

1 Introduction

This section should introduce the problem of lossless data compression and the objectives of this project.

2 Background

2.1 Information Theory

2.2 Transform Coding

2.3 Statistical Modeling

2.4 Entropy Coding

3 Methodology

3.1 System Architecture

3.2 Algorithm Selection

3.3 Optimization Strategy

3.4 Benchmarking

4 Implementation

4.1 Burrows-Wheeler Transform

4.2 Move-to-Front and Run-Length Encoding

4.3 PPM Model

4.4 Entropy Coders

4.5 Parallel Processing

4.6 Performance Optimizations

5 Results

5.1 Compression Ratio Evolution

5.2 Benchmark Comparison

Table 1: Compression Ratio Comparison (TODO: fill with actual data)

File	Size	gzip	bzip2	xz	zstd	G07 v5.0
A	1.3MB	?	54.05%	?	?	53.30%

5.3 Performance Analysis

5.4 Ablation Studies

6 Conclusion

6.1 Summary of Achievements

6.2 Lessons Learned

6.3 Limitations

6.4 Future Work